

MOST IMPORTANT QUESTIONS BIOLOGY

Expert Analysis of Board Paper Sets 57/1/1, 57/5/1, 57/7/1

Created for Thunderstudy By Pragya

Unit VI: Reproduction

Chapter 2: Sexual Reproduction in Flowering Plants

- **Pollination & Co-evolution:** Explain the mutualistic relationship between the Fig tree and the Wasp, or the Orchid Ophrys and the bee. How do plants ensure cross-pollination (Outbreeding devices)?
- **Embryo & Endosperm:** Why does endosperm development precede embryo development? Differentiate between the structure of a monocot and a dicot embryo. **Diagram-Based**
- **Gametophyte Development:** Describe the development of the male gametophyte (pollen) and the female gametophyte (embryo sac). Identify synergids, antipodals, and filiform apparatus. **Diagram-Based**
- **Apomixis & Polyembryony:** Importance in hybrid seed industry; difference from parthenocarpy.

Chapter 3: Human Reproduction

- **Menstrual Cycle:** Events in ovary and uterus during follicular/luteal phases. Graphical representation of LH, FSH, Estrogen, and Progesterone levels. **Graph-Based**
- **Gametogenesis:** Spermatogenesis and Oogenesis schematic representations. Role of FSH and LH.
- **Fertilization & Implantation:** Acrosomal reaction and the cortical reaction (polyspermy block). Formation and function of the placenta.
- **Labelled Diagrams:** Human sperm, seminiferous tubule sectional view, and ovary sectional view. **Must Practice**

Chapter 4: Reproductive Health

- **Assisted Reproductive Technologies (ART):** Expand and explain IVF, ZIFT, GIFT, ICSI, and IUI. Recommendations for each.
- **Contraception:** Functioning of oral contraceptive pills (Saheli) and IUDs (CuT, LNG-20).
- **Amniocentesis:** Applications and the reasons for the statutory ban.

Unit VII: Genetics and Evolution

Chapter 5: Principles of Inheritance and Variation

- **Genetic Disorders:** Chromosomal basis and symptoms of Down's, Turner's, and Klinefelter's syndromes.
- **Pedigree Analysis:** Analysis for Haemophilia, Colour blindness, Sickle cell anaemia, and Thalassemia.
- **Inheritance Patterns:** Co-dominance (ABO blood grouping), Multiple Alleles, Polygenic inheritance, and Pleiotropy.
- **Sex Determination:** Birds (ZZ-ZW), Humans (XX-XY), and Honeybees (Haplodiploidy).

Chapter 6: Molecular Basis of Inheritance

- **DNA Replication:** Machinery (Helicase, Polymerase, Ligase) and replication fork formation (Okazaki fragments). **Diagram-Based**
- **Transcription & Translation:** Processing of hnRNA (Splicing, Capping, Tailing) and the Lac Operon model.
- **Experiments:** Hershey-Chase experiment (Genetic material) and Meselson-Stahl experiment (Semi-conservative replication).
- **DNA Fingerprinting:** Technical steps and forensic/legal applications.

Chapter 7: Evolution

- **Mechanisms:** Homologous vs Analogous organs (Convergent vs Divergent evolution).
- **Natural Selection:** Industrial Melanism (Peppered moth study).
- **Adaptive Radiation:** Darwin's Finches and Australian Marsupials.
- **Hardy-Weinberg Principle:** Statement and factors affecting equilibrium.

Unit VIII: Biology in Human Welfare

Chapter 8: Human Health and Disease

- **Immunity:** Innate vs Acquired, Active vs Passive. Structure of an antibody molecule. **Diagram-Based**
- **Diseases:** Life cycle of Plasmodium. Typhoid, Pneumonia, Ascariasis, and Ringworm.
- **AIDS & Cancer:** Retrovirus replication (HIV). Benign vs Malignant tumors (Metastasis).
- **Drugs:** Source and effects of Opioids (Heroin), Cannabinoids, and Coca alkaloids.

Chapter 10: Microbes in Human Welfare

- **Sewage Treatment:** Primary and Secondary treatment (BOD, flocs, activated sludge).
- **Bioactive Molecules:** Microbes for Cyclosporin A, Statins, and Streptokinase.
- **Biofertilizers:** Role of Mycorrhiza and Cyanobacteria.

Unit IX: Biotechnology

Chapter 11: Principles and Processes

- **rDNA Tools:** Restriction Endonucleases naming (EcoRI). Palindromes and Sticky ends.
- **Cloning Vectors:** Structure of **pBR322**. Role of selectable markers and Insertional Inactivation. **Diagram-Based**

- **Processes:** PCR steps (Denaturation, Annealing, Extension) and Bioreactors.

Chapter 12: Biotechnology and its Applications

- **Agriculture:** Bt Cotton development (Cry genes) and RNA interference (RNAi) in Tobacco plants.
- **Medicine:** Genetically engineered Insulin production and Gene Therapy (ADA deficiency).
- **Transgenic Animals:** Reasons for production (Rosie the cow, Vaccine safety).

Unit X: Ecology

Chapter 13: Organisms and Populations

- **Population Growth:** Exponential vs Logistic curves. Verhulst–Pearl equation explanation. **Graph-Based**
- **Interactions:** Mutualism, Parasitism, Commensalism, and Competition with examples.

Chapter 14: Ecosystem

- **Pyramids:** Energy pyramid (always upright) vs Inverted pyramids (Biomass in aquatic).
- **Productivity:** Differentiate between GPP and NPP.

Chapter 15: Biodiversity and Conservation

- **Loss of Biodiversity:** The "Evil Quartet" (Habitat loss, Alien species like Nile Perch, etc.).
- **Species–Area Relationship:** Humboldt's observation and equation ($\log S = \log C + Z \log A$).
- **Conservation:** In-situ vs Ex-situ (Sacred groves, Hotspots vs Seed banks).

Most Important Questions Biology By Pragya Thunderstudy
Success Guide for Biology Boards